School Worksheet

Class 12 Mathematics, Chapter 13: Probability

Maximum marks: **51** **20** questions

Name _____ Date _____ Score _____ / 51

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ — 1 mark

If $P(A) = 0.8$, $P(B) = 0.5$ and $P(B|A) = 0.4$, then $P(A \cap B)$ is:

(A) 0.32 **(B)** 0.25

(C) 0.1 **(D)** 0.4

2 MCQ — 1 mark

The probability that a particular item is available in three shops A, B and C is $\frac{4}{5}$, $\frac{3}{4}$ and $\frac{2}{3}$ respectively. If a person visits all three shops, the probability that the item is available in at least one of them is:

(A) $\frac{59}{60}$

(B) 1

(C) $\frac{1}{60}$

(D) $\frac{9}{60}$

3 MCQ — 1 mark

A die is thrown. If E is the event “the number is 4” and F is the event “the number is even”, then $P(E|F)$ is:

(A) $\frac{1}{6}$

(B) $\frac{1}{2}$

(C) $\frac{1}{3}$

(D) $\frac{2}{3}$

Q No.	Question	Marks
4	MCQ If $P(A) = \frac{1}{2}$ and $P(B) = 0$, then $P(A\|B)$ is: **(A)** 0 **(B)** $\frac{1}{2}$ **(C)** 1 **(D)** not defined	1 mark
5	MCQ A random variable X has the distribution $P(X = 0) = k$, $P(X = 1) = 2k$, $P(X = 2) = 3k$. The value of k is: **(A)** $\frac{1}{6}$ **(B)** $\frac{1}{3}$ **(C)** $\frac{1}{2}$ **(D)** 1	1 mark
6	Assertion-Reason **Assertion (A):** Events A and B with $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$ and $P(A \cap B) = \frac{1}{6}$ are independent. **Reason (R):** Two events A and B are independent if and only if $P(A \cap B) = P(A)\,P(B)$. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer Two cards are drawn one after the other without replacement from a well-shuffled pack of 52 cards. Find the probability that both are kings.	2 marks

Level 2

18 marks

8	Short answer The chances of P, Q and R getting selected as CEO of a company are in the ratio $4 : 1 : 2$. The probabilities that the company increases its profits under the new CEO P, Q or R are 0.3, 0.8 and 0.5 respectively. If the company increased its profits, find the probability that it is due to the appointment of R as CEO.	3 marks
9	Short answer Find the probability distribution of the number of heads in two tosses of a fair coin. Also find its mean.	2 marks

Q No.	Question	Marks
10	Short answer Given $P(A) = 0.6$, $P(B) = 0.3$ and $P(A \cap B) = 0.2$, find $P(A\|B)$, $P(B\|A)$ and $P(A'\|B')$.	3 marks
11	Short answer A and B independently try to solve a problem with probabilities $\frac{1}{2}$ and $\frac{1}{3}$ respectively. Find the probability that (a) the problem is solved (b) exactly one of them solves it.	3 marks
12	Short answer A family has two children. Find the probability that both are boys, given that at least one of them is a boy.	2 marks
13	Short answer A random variable X has $P(X = x) = kx$ for $x = 1, 2, 3, 4$ and 0 otherwise. Find k, $P(X \geq 3)$ and the mean of X.	3 marks
14	Short answer Three coins are tossed. Let E: "three heads or three tails" and F: "at least two heads". Are E and F independent?	2 marks
Level 3		**25 marks**
15	Long answer A factory has machines A, B and C producing 50%, 30% and 20% of the items, of which 2%, 3% and 4% respectively are defective. An item picked at random is found defective. Find the probability that it was produced by machine A.	5 marks
16	Short answer In answering a multiple-choice question (4 options), a student either knows the answer or guesses. The probability that she knows the answer is $\frac{3}{4}$. If she answers correctly, find the probability that she knew the answer.	3 marks
17	Long answer Two cards are drawn successively without replacement from a well-shuffled pack of 52 cards. Find the probability distribution of the number of aces and its mean.	5 marks
18	Short answer A random variable X has the distribution: X: $0, 1, 2, 3, 4, 5, 6, 7$ with $P(X)$: $0,\ k,\ 2k,\ 2k,\ 3k,\ k^2,\ 2k^2,\ 7k^2 + k$. Find k, $P(X < 3)$ and $P(X > 6)$.	3 marks
19	Long answer A laboratory blood test is 99% effective in detecting a disease when it is present, but gives a false positive for 0.5% of healthy persons. If 0.1% of the population has the disease, find the probability that a person who tests positive actually has the disease.	5 marks

Q No.	Question	Marks
20	Case study **Case Study:** An NGO organises a charity event to distribute woollen caps to children in winter. The caps are in three boxes: Box I has 30 red caps, Box II has 20 red and 10 green caps, and Box III has 30 green caps. The probability that Box i is selected and a cap picked out is $\frac{i}{6}$, $i = 1, 2, 3$. A person selects a cap. **(i)** What is the probability that he selects a red cap? 2 marks **(ii)** If he selects a green cap, what is the probability that it has come from Box II? 2 marks	4 marks

Answer key

Chapter 13: Probability, School Worksheet. Full step-wise solutions are in the solutions sheet.

Level 1

1	(A) 0.32
2	(A) $\frac{59}{60}$
3	(C) $\frac{1}{3}$
4	(D) not defined
5	(A) $\frac{1}{6}$
6	(A) Both A and R are true and R is the correct explanation of A
7	$\frac{1}{221}$

Level 2

8	$P(E_3 \mid A) = \frac{1}{3}$
9	X: $0, 1, 2$ with $P(X)$: $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$; mean $= 1$
10	$P(A \mid B) = \frac{2}{3}$, $P(B \mid A) = \frac{1}{3}$, $P(A' \mid B') = \frac{3}{7}$
11	(a) $\frac{2}{3}$ (b) $\frac{1}{2}$
12	$\frac{1}{3}$
13	$k = \frac{1}{10}$, $P(X \geq 3) = \frac{7}{10}$, mean $= 3$
14	Yes, $P(E \cap F) = \frac{1}{8} = P(E)\,P(F)$, so E and F are independent

Level 3

15	$P(E_1 \mid D) = \frac{10}{27}$
16	$P(E_1 \mid A) = \frac{12}{13}$
17	X: $0, 1, 2$ with $P(X)$: $\frac{188}{221}, \frac{32}{221}, \frac{1}{221}$; mean $= \frac{2}{13}$
18	$k = \frac{1}{10}$, $P(X < 3) = \frac{3}{10}$, $P(X > 6) = \frac{17}{100}$
19	$P(E_1 \mid A) = \frac{22}{133}$
20	(i) $\frac{7}{18}$ (ii) $\frac{2}{11}$